

Harshavardhana Gowda

☎ (+1) 530 760 8789 | ✉ tgharshavardhana@gmail.com | [in](#) LinkedIn |

Education

University of California

PhD in Electrical and Computer Engineering *GPA: 3.83/4.0*

Davis, USA

Sep 2022 – Present

University of California

M. S. in Electrical and Computer Engineering *GPA: 3.97/4.0*

Davis, USA

Sep 2022 – Aug 2024

Indian Institute of Space Science and Technology

B.Tech in Avionics *GPA: 8.1/10*

India

Aug 2014 – May 2018

Publications

[emg2speech: synthesizing speech from electromyography using self-supervised speech models](#)

<https://harshavardhanatg.github.io/emg2speech.github.io/>

Harshavardhana T. Gowda*, Daniel C. Comstock, Lee M. Miller.

Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2026.

[Non-invasive electromyographic speech neuroprosthesis: a geometric perspective](#)

<https://harshavardhanatg.github.io/emg2text.github.io/>

Harshavardhana T. Gowda*, Lee M. Miller.

Findings of the Association for Computational Linguistics: ACL 2026.

NeurIPS 2025 TS4H Workshop (Spotlight).

[Geometry of orofacial neuromuscular signals: speech articulation decoding using surface electromyography](#)

<https://harshavardhanatg.github.io/orofacialEMG.github.io/>

Harshavardhana T. Gowda*, Zachary McNaughton, Lee M. Miller.

Journal of Neural Engineering, 2025.

[Topology of surface electromyogram signals: hand gesture decoding on Riemannian manifolds](#)

<https://harshavardhanatg.github.io/handEMG.github.io/>

Harshavardhana T. Gowda*, Lee M. Miller.

Journal of Neural Engineering, 2024.

[A database of upper limb surface electromyogram signals from demographically diverse individuals](#)

<https://harshavardhanatg.github.io/handEMG.github.io/>

Harshavardhana T. Gowda*, Neha Kaul, Carlos Carrasco, Marcus A. Batraw, Safa Amer, Saniya Kotwal, Selena Lam, Zachary McNaughton, Ferdous Rahimi, Sana Shehabi, Jonathon S. Schofield, Lee M. Miller.

Scientific Data, 2025.

Research Focus

University of California, Davis, USA

Graduate Researcher

Sep 2022 – May 2027

- I work on building end-to-end, 0-to-1 systems at the intersection of human sensing and machine intelligence. My research focuses on inferring the neuromotor, visual, and kinematic intent underlying human actions and using these signals to build predictive systems that augment human interaction

with physical and virtual environments. I work across the full technical stack required to translate research ideas into functioning systems, including experimental design, large-scale human-subject data collection, algorithm and model development for multimodal time-series and perception data, and deployment in user-facing prototypes.

Ongoing Projects

Multimodal speech decoding using EMG, video, and residual audio for restoring speech in individuals with neuromotor disorders

I am leading the development of a non-invasive EMG-based speech neuroprosthesis for individuals with amyotrophic lateral sclerosis (ALS) and laryngectomy. The system infers phonetic structure directly from orofacial EMG and video recorded during silent speech articulation, then synthesizes intelligible audible speech. I am scaling our prior work by training foundation models on hundreds of hours of large-scale EMG data, with the goal of building a robust, generalizable, and deployable communication interface for people who have lost the ability to speak.

Understanding human dexterity from multimodal sensing to advance robotics

I am characterizing fine-grained dexterous manipulation in the human hand using complementary sensing modalities. This includes EMG to measure muscle activation associated with force production and stabilization, video to estimate hand and hand-object kinematics and interaction states, and eye tracking to quantify gaze allocation and its temporal coupling to manipulation subgoals. I am focusing on executing large-scale multimodal experiments and developing learning models that jointly understand muscle, kinematic, and gaze signals during contact-rich manipulation. This work aims to provide mechanistic insight into how humans coordinate sensing and control to achieve stable, dexterous manipulation, informing the design of more capable robotic and humanoid systems as well as improved interaction models in virtual and augmented reality.

Skills

Programming: Python, C, CUDA.

Frameworks: PyTorch

Professional Work Experience

Indian Space Research Organization

Satellite Design Engineer

India

Sep 2018 – June 2022

- Contributed to communication system design of Geosats including GSAT-20 and GISAT (EOS series).
- Contributed to power system design and electronics bus integration of human space mission crew module.
- Developed a software platform to aid the design of digital beamforming to allow coexistence of LTE and MSS. Tested the system in an anechoic chamber using S-band receive antenna. (*Member, Research Team - Novel Technologies*)
- Developed a platform for Digital Video Broadcasting Satellite Second Generation (DVB - S2) standard. Modeled RF satellite systems, communication signals, communication channels and impairments. Analyzed the performance of satellite transponder by measuring error vector magnitude for various modulation schemes. (*Member, Communication Systems Development Team*)
- Conducted failure mode, effects and criticality analysis, and worst-case circuit stress analysis for satellite electronics. Devised experimental investigation techniques to scrutinize fabrication and design errors when circuit performance does not conform to specifications in emulated space environment.

Awards & Achievements

Neuralstorm Fellowship, NSF NRT Award No. 2152260 and Ellis Fund, UC Davis, 2024
Dollars 66,551 for conducting research in neural engineering.

UC Davis Summer Fellowship Dollars 24,000 for conducting research in engineering.

Department of Space Scholarship, Indian Space Research Organization, 2014
Dollars 10,000 to support undergraduate studies in space research.